

Rising Waters - Flood Prediction System

Project Overview

Rising Waters is a Machine Learning + Flask web application that predicts whether a flood will occur based on environmental and rainfall parameters.

The system analyzes weather conditions like temperature, humidity, rainfall distribution, and seasonal data to classify:

-  Flood
 -  No Flood
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Machine Learning Model

- Algorithm: Random Forest Classifier
 - Preprocessing: StandardScaler
 - Input Features: 10 weather parameters
 - Output: Binary classification (0 = No Flood, 1 = Flood)
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Tech Stack

- Python 
 - Flask 
 - Scikit-learn 
 - Pandas 
 - HTML, CSS 
 - Joblib (model saving)
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Input Features

1. Temp
2. Humidity

3. Cloud Cover
 4. ANNUAL Rainfall
 5. Jan-Feb Rainfall
 6. Mar-May Rainfall
 7. Jun-Sep Rainfall
 8. Oct-Dec Rainfall
 9. Avg June
 10. Sub value
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How It Works

1. User enters weather values in UI
 2. Data is scaled using StandardScaler
 3. Model predicts flood probability
 4. Output is displayed on web page
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Run Locally

```
pip install -r requirements.txt
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```
python model_training.py
```

```
python app.py
```